

Scenario: Management of adults with type 1 diabetes

Last revised in December 2024

From age 18 years onwards.

What initial care and support should I offer an adult with diabetes?

- Ensure that an individual care plan (</topics/diabetes-type-1/management/management-adults/#individual-care-plan>) is set up for the person by the diabetes specialist team.
 - After the initial plan is agreed, put arrangements in place to implement it without inappropriate delay. Arrange a review to reassess the person in the following weeks.
 - Review the plan annually and modify it if necessary, taking into account changes in the person's wishes, circumstances, and clinical findings, and record the details.
 - Consider using the Scottish Intercollegiate Guidelines Network (SIGN) toolkit Optimising glycaemic control in people with type 1 diabetes (<https://www.sign.ac.uk/our-guidelines/optimising-glycaemic-control-in-people-with-type-1-diabetes/>) to support people with type 1 diabetes.
- Offer a structured education programme of proven benefit, for example the DAFNE (<https://dafne.nhs.uk/>) (dose-adjustment for normal eating) programme.
 - Offer this programme 6–12 months after diagnosis or at any time that is clinically appropriate and suitable for the person, regardless of duration of type 1 diabetes.
 - Explain to the person and/or their family/carers that structured education is an integral part of diabetes care.
 - For people who are unable or unwilling to participate in group education, provide an alternative of equal standard.
- Provide information on:
 - The means and frequency of communicating with the diabetes specialist team. Ensure that the person and/or their family/carers know how to contact the diabetes team during working hours and out of hours, as available.
 - Government disability benefits — Benefits calculators (<https://www.gov.uk/benefits-calculators>) are available at www.gov.uk (<https://www.gov.uk>).
- Manage lifestyle issues, such as diet, exercise, and alcohol intake.
 - See the section on Lifestyle management (</topics/diabetes-type-1/management/management-adults/#lifestyle-management>) for more information.
- Provide up-to-date information on diabetes support groups (local and national), including information on how to contact them and the benefits of membership.
 - For example, the Diabetes UK website (www.diabetes.org.uk) (<https://www.diabetes.org.uk/>) has information on Living with diabetes (<https://www.diabetes.org.uk/guide-to-diabetes>) and a support helpline (https://www.diabetes.org.uk/how_we_help/helpline) for people with diabetes and their family/carers.

- Provide this information at the time of diagnosis and periodically thereafter. Written information should also be available from the diabetes team.

Individual care plan

- **At the time of diagnosis, the diabetes specialist team should develop with (and explain to) all adults with type 1 diabetes a plan for their early care.** To agree such a plan will generally require:
 - Medical assessment to:
 - Confirm diagnosis of type 1 diabetes.
 - Ensure appropriate acute care is given when needed.
 - Review and detect potentially confounding diseases and drugs.
 - Detect adverse vascular risk factors, such as hypertension.
 - Environmental assessment to understand:
 - The social, home, work, and recreational circumstances of the person and carers.
 - The person's preferences in nutrition and physical activity.
 - Other relevant factors, such as substance use.
 - Cultural and educational assessment to identify prior knowledge and to enable optimal advice and planning on:
 - Treatment modalities.
 - Diabetes education programmes.
 - Assessment of emotional state to determine the appropriate pace of education.
- **The initial diabetes assessment will include a general examination and a review of the following:**
 - Acute medical history.
 - Social, cultural, and educational history/lifestyle review.
 - Complications history/symptoms.
 - Long-term/recent diabetes history.
 - Other medical history/systems.
 - Family history of diabetes/cardiovascular disease.
 - Medication history/current medicines.
 - Vascular risk factors.
 - Smoking status, general examination, weight/body mass index.
 - Foot/eye/vision examination.
 - Urine albumin excretion/urine protein/serum creatinine.
 - Psychological wellbeing.
 - Attitudes to medicine and self-care.
 - Immediate family and social relationships and availability of informal support.
- **The results of the assessment should be used to agree an individual and culturally appropriate care plan for the person with type 1 diabetes, which should include aspects of the following:**
 - Diabetes education (sites and timescales of diabetes education and nutritional advice).
 - Insulin therapy (insulin injections, regimens, and dose adjustment).
 - Self-monitoring of blood glucose.
 - Treatment targets.
 - Hypoglycaemia (symptoms, risks, and treatment).
 - Management of special situations (for example driving, fasting, and physical activity).
 - Management of diabetes during a period of illness.

- Cardiovascular risk factors (monitoring and management).
- Complications (monitoring and management).
- Means and frequency of communicating with the diabetes specialist team.
- Frequency and content of follow-up consultations, including review of HbA1c levels and experience of hypoglycaemia, and next annual review.
- Contraception and pregnancy planning advice (when appropriate).

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Insulin therapy for adults with type 1 diabetes

- Insulin therapy should only be initiated and managed by healthcare professionals with the relevant expertise and training.
- See the CKS topic on [Insulin therapy in type 1 diabetes \(/topics/insulin-therapy-in-type-1-diabetes/\)](/topics/insulin-therapy-in-type-1-diabetes/) for detailed information on insulin therapy, including available insulin preparations, recommended regimens, information on injection technique and injection sites, storage of insulin, adverse effects, safety issues, and general information on lifestyle and cultural aspects of insulin treatment (such as driving whilst on insulin treatment, fasting, shift work, holiday and travel, and insurance).

What is the target HbA1c for adults with type 1 diabetes

- Support adults with type 1 diabetes to aim for a target HbA1c level of 48 mmol/mol (6.5%) or lower, to minimize the risk of long-term vascular complications (</topics/diabetes-type-1/background-information/complications/>).
- For people with type 1 diabetes and chronic kidney disease not treated with dialysis, aim for a target HbA1C of less than 6.5% to less than 8%.
- Agree an individualized HbA1c target with the person, taking into account factors such as their daily activities, aspirations, likelihood of complications, comorbidities, occupation, and history of hypoglycaemia (blood glucose levels less than 3.5 mmol/L).
- Ensure that aiming for an HbA1c target is not accompanied by problematic hypoglycaemia.
- Measure HbA1c levels every 3–6 months in most people.
 - Consider measuring HbA1c levels more often if the person's blood glucose control is suspected to be changing rapidly, for example if the HbA1c level has risen unexpectedly above a previously sustained target.
 - Inform the person of their results after each measurement, and ensure that their most recent result is available at the time of consultation.

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What should I advise on self-monitoring of blood glucose

- Self-monitoring of blood glucose refers to home blood glucose testing for people with diabetes to determine the amount of glucose in the blood at any given time.

- **Continuous glucose monitoring (CGM) is a way of measuring glucose levels continuously throughout the day and night via a tiny electrode inserted under the skin.**
 - Continuous glucose monitoring systems provide continuous measurements (every 1–5 minutes) of glucose concentrations in the interstitial fluid, which correlate with blood glucose levels. It can provide both retrospective and real-time information to detect hypoglycaemic and hyperglycaemic episodes, predict impending hypoglycaemia, and detect wide fluctuations in glucose levels (glycaemic variability).
 - A choice of real-time continuous glucose monitoring (rtCGM) or intermittently scanned continuous glucose monitoring (isCGM) should be offered to all adults with type 1 diabetes. If the person cannot use or does not want rtCGM or isCGM, capillary blood glucose monitoring should be offered.
 - CGM should be provided by a team with expertise in its use, as part of supporting people to self-manage their diabetes.
 - Advise adults with type 1 diabetes who are using CGM that they will still need to take capillary blood glucose measurements (although they can do this less often). Explain that this is because:
 - They will need to use capillary blood glucose measurements to check the accuracy of their CGM device
 - They will need capillary blood glucose monitoring as a back-up (for example, when their blood glucose levels are changing quickly or if the device stops working).
 - The person's use of CGM should be monitored and reviewed as part of reviewing their diabetes care plan.
- All people with type 1 diabetes will need a blood glucose monitor, lancets (which fit into a finger-pricking device), and testing strips (on which the blood is collected for testing in the machine) to monitor capillary blood glucose.
 - See the section on [Target glucose self-monitoring levels \(/topics/diabetes-type-1/management/management-adults/#target-glucose-self-monitoring-levels\)](#) for information on blood glucose self-monitoring target levels.
- **Self-monitoring skills should be taught at the time of diagnosis and the start of insulin therapy, and should be reviewed at least annually.** The review should include checking that the person knows:
 - How to use their blood glucose monitoring meter — this can be done by asking them to demonstrate their technique.
 - When to test — for example before breakfast and 2 hours after meals, during periods of illness, before driving, and if they feel hypoglycaemic.
 - How to assess and respond to test results, including how to:
 - Interpret trends in blood glucose levels as well as individual readings — this is important for adjusting insulin doses.
 - Identify the causes of high and low blood glucose values.

What are the target glucose self-monitoring levels for adults with type 1 diabetes?

- **Advise adults who are using capillary blood glucose monitoring to routinely monitor their blood glucose levels and to measure at least 4 times a day (including before meals and before bed).**
 - More frequent monitoring (up to 10 times a day or more) may be required if any of the following apply:
 - The [target HbA1c level \(/topics/diabetes-type-1/management/management-adults/#target-hba1c\)](#) is not achieved.
 - The frequency of hypoglycaemic episodes increases.

- There is a legal requirement to do so, such as before driving.
 - During periods of illness (</topics/diabetes-type-1/management/management-adults/#managing-intercurrent-illness>).
 - Before, during, and after sport.
 - When planning a pregnancy, during pregnancy, and while breastfeeding. See the National Institute for Health and Care Excellence (NICE) guideline on [Diabetes in pregnancy \(https://www.nice.org.uk/guidance/ng3\)](https://www.nice.org.uk/guidance/ng3) for more information.
 - If there is a need to know blood glucose levels more than 4 times a day for other reasons, for example in people with impaired awareness of hypoglycaemia. See the section on [Impaired awareness of hypoglycaemia \(/topics/insulin-therapy-in-type-1-diabetes/management/insulin-therapy-type-1-diabetes/#impaired-awareness-of-hypoglycaemia\)](/topics/insulin-therapy-in-type-1-diabetes/management/insulin-therapy-type-1-diabetes/#impaired-awareness-of-hypoglycaemia) in the CKS topic on [Insulin therapy in type 1 diabetes \(/topics/insulin-therapy-in-type-1-diabetes/\)](/topics/insulin-therapy-in-type-1-diabetes/) for more information.
 - If it is necessary because of the person's lifestyle, for example they drive for a long period of time, undertake a high-risk activity or occupation, or travel frequently or across time zones.
- **Discuss optimal targets for glucose self-monitoring.**
 - **The optimal targets for glucose self-monitoring in adults with type 1 diabetes are:**
 - Fasting plasma glucose level of 5–7 mmol/L on waking.
 - Plasma glucose level of 4–7 mmol/L before meals at other times of the day.
 - For adults who choose to test after meals, plasma glucose level of 5–9 mmol/L at least 90 minutes after eating.
 - **Agree bedtime target plasma glucose levels with the person. This should:**
 - Take into account the timing of the last meal and its related insulin dose.
 - Be consistent with the recommended fasting level on waking.
 - **Remind the person of the need to be particularly careful to avoid hypoglycaemia when driving.** See the section on [Driving \(/topics/insulin-therapy-in-type-1-diabetes/management/insulin-therapy-type-1-diabetes/#driving\)](/topics/insulin-therapy-in-type-1-diabetes/management/insulin-therapy-type-1-diabetes/#driving) in the CKS topic on [Insulin therapy in type 1 diabetes \(/topics/insulin-therapy-in-type-1-diabetes/\)](/topics/insulin-therapy-in-type-1-diabetes/) for more information.

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Lifestyle management

- See the sections on [Diet \(/topics/diabetes-type-1/management/management-adults/#diet\)](/topics/diabetes-type-1/management/management-adults/#diet), [Exercise \(/topics/diabetes-type-1/management/management-adults/#exercise-physical-activity\)](/topics/diabetes-type-1/management/management-adults/#exercise-physical-activity), [Alcohol intake \(/topics/diabetes-type-1/management/management-adults/#alcohol-intake\)](/topics/diabetes-type-1/management/management-adults/#alcohol-intake), and [Substance misuse \(/topics/diabetes-type-1/management/management-adults/#smoking-substance-misuse\)](/topics/diabetes-type-1/management/management-adults/#smoking-substance-misuse) for information on managing these lifestyle issues.
- Information on managing other lifestyle and cultural issues, such as driving, fasting, holiday and travel, and shift work, are available in the section on [Additional information and advice \(/topics/insulin-therapy-in-type-1-diabetes/management/insulin-therapy-type-1-diabetes/#lifestyle-cultural-issues\)](/topics/insulin-therapy-in-type-1-diabetes/management/insulin-therapy-type-1-diabetes/#lifestyle-cultural-issues) in the CKS topic on [Insulin therapy in type 1 diabetes \(/topics/insulin-therapy-in-type-1-diabetes/\)](/topics/insulin-therapy-in-type-1-diabetes/).

Advice on diet

- **Provide information on the effects of different foods and drinks on blood glucose control.**
 - Do not advise a low glycaemic index diet for blood glucose control.
 - Offer carbohydrate-counting training (a meal planning technique for managing blood glucose levels by matching carbohydrate quantities to insulin doses) as part of the [structured education programme \(/topics/diabetes-type-1/management/management-adults/#initial-care-support\)](#) for self-management.
 - Consider carbohydrate-counting courses for people who are waiting for a more detailed structured education programme or are unable to take part in a stand-alone structured education programme.
 - Refer the person to the specialist diabetes team (or make programmes available) to enable them to make optimal choices about the variety of foods they wish to consume and on appropriate insulin dose changes when taking different quantities of those foods.
 - The Diabetes UK website (www.diabetes.org.uk (<https://www.diabetes.org.uk/>)) has dietary information for people with type 1 diabetes, including information on [carbohydrates and diabetes](#) (<https://www.diabetes.org.uk/Guide-to-diabetes/Enjoy-food/Carbohydrates-and-diabetes/>), what to [eat](#) (<https://www.diabetes.org.uk/guide-to-diabetes/enjoy-food/eating-with-diabetes/i-have-type-1-diabetes>) and [drink](#) (<https://www.diabetes.org.uk/Guide-to-diabetes/Enjoy-food/what-to-drink-with-diabetes/>), [food shopping tips](#) (<https://www.diabetes.org.uk/Guide-to-diabetes/Enjoy-food/Food-shopping-for-diabetes/>), and [how to cut down on sugar](#) (<https://www.diabetes.org.uk/Guide-to-diabetes/Enjoy-food/Carbohydrates-and-diabetes/how-to-cut-down-on-sugar/>).
- **Provide dietary information for the prevention of cardiovascular disease (CVD).**
 - Advise adults with type 1 diabetes on the importance of a healthy, balanced diet in reducing the risk of CVD. The diet should be low in fat, sugar, and salt, and contain at least 5 portions of fruit and vegetables a day.
 - If necessary, offer dietetic support (or advise the person to contact their specialist diabetes team) to help optimize body weight and blood glucose control.
- **Modify nutritional recommendations to take account of associated features of diabetes, including:**
 - Excess weight and obesity. See the CKS topic on [Obesity \(/topics/obesity/\)](#) for management information.
 - Being underweight.
 - Eating disorders. See the CKS topic on [Eating disorders \(/topics/eating-disorders/\)](#) for management information.
 - Hypertension. See the section on [Antihypertensive treatment \(/topics/diabetes-type-1/management/management-adults/#antihypertensive-treatment\)](#) for management information.
 - Chronic kidney disease (CKD). See the section on [Diabetic kidney disease \(/topics/diabetes-type-1/management/management-adults/#diabetic-kidney-disease\)](#) for management information.
- **Be aware of appropriate nutritional advice on common topics of concern and interest to adults with type 1 diabetes.**
 - These may include:
 - Body weight, energy balance, and obesity management.
 - Cultural and religious diets, feasts, and fasts.
 - Diabetic foods and sweeteners.
 - Dietary fibre and protein intake.
 - Vitamin and mineral supplements.

- Alcohol intake.
- Matching carbohydrate, insulin, and physical activity.
- Salt intake in hypertension.
- Comorbidities, including nephropathy and renal failure, coeliac disease, cystic fibrosis, or eating disorders.
- Use of peer support groups.
- Refer the person to a dietitian or the specialist diabetes team for information on these and other dietary issues.

Advice on alcohol intake

- **Advise the person with type 1 diabetes and/or their family/carers:**
 - To avoid drinking alcohol on an empty stomach as the alcohol will be absorbed faster. They should:
 - Eat a snack that contains carbohydrate (such as a sandwich or crisps) before and after drinking alcohol. Extra insulin is not required.
 - Measure their blood glucose regularly, and maintain their blood glucose with appropriate carbohydrate intake.
 - That alcohol may exacerbate or prolong the hypoglycaemic effect of insulin, including nocturnal hypoglycaemia.
 - Signs of hypoglycaemia may become less clear, and delayed hypoglycaemia may occur (which can be hours after alcohol consumption).
 - See the section on [hypoglycaemia \(/topics/insulin-therapy-in-type-1-diabetes/management/insulin-therapy-type-1-diabetes/#managing-hypoglycaemia\)](/topics/insulin-therapy-in-type-1-diabetes/management/insulin-therapy-type-1-diabetes/#managing-hypoglycaemia) in the CKS topic on [Insulin therapy in type 1 diabetes \(/topics/insulin-therapy-in-type-1-diabetes/\)](/topics/insulin-therapy-in-type-1-diabetes/) for detailed information on hypoglycaemia.
 - To always wear some form of diabetes identification as the reduced awareness of hypoglycemia may be confused with alcohol intoxication. Examples of diabetes identification include:
 - A MedicAlert® bracelet, necklace, or watch (available from www.medicalert.org.uk (<http://www.medicalert.org.uk/>)).
 - Diabetes ID Card (available from www.diabetes.co.uk (<https://www.diabetes.co.uk/diabetic-products/medipal-card.html>)).
- **Give information on the recommended maximum alcohol intake.**
 - The [Chief Medical Officers' guideline](http://assets.publishing.service.gov.uk/government/uploads/system/uploads/attachment_data/file/545739/GDG_report-Jan2016.pdf) (http://assets.publishing.service.gov.uk/government/uploads/system/uploads/attachment_data/file/545739/GDG_report-Jan2016.pdf) for both men and women states that:
 - It is safest not to regularly drink more than 14 units of alcohol per week, to keep health risks from drinking alcohol to a low level.
 - People who drink as much as 14 units of alcohol per week should spread this evenly over 3 days or more. One or two heavy drinking sessions increases the risk of death from long-term illnesses and from accidents and injuries.
 - The risk of developing a range of illnesses (such as cancers of the mouth, throat, and breast) increases with any amount of alcohol consumed on a regular basis.
 - If people wish to cut down the amount of alcohol they are drinking, a good way to help achieve this is to have several drink-free days each week.
 - For women who are pregnant or planning a pregnancy, the safest approach is not to drink alcohol at all, to keep risks to the baby to a minimum.
 - See the section on [Unit of alcohol \(/topics/alcohol-problem-drinking/background-information/definition/#unit-of-alcohol\)](/topics/alcohol-problem-drinking/background-information/definition/#unit-of-alcohol) in the CKS topic on [Alcohol - problem](#)

[drinking](#) ([/topics/alcohol-problem-drinking/](#)) for more information on alcohol units.

- Information on [Alcohol and diabetes](https://www.diabetes.org.uk/Guide-to-diabetes/Enjoy-food/what-to-drink-with-diabetes/Alcohol-and-diabetes) (<https://www.diabetes.org.uk/Guide-to-diabetes/Enjoy-food/what-to-drink-with-diabetes/Alcohol-and-diabetes>) and [Type 1 diabetes and drinking](https://www.diabetes.org.uk/guide-to-diabetes/young-adults/type-1-drinking) (<https://www.diabetes.org.uk/guide-to-diabetes/young-adults/type-1-drinking>) are also available on the Diabetes UK website (www.diabetes.org.uk) (<https://www.diabetes.org.uk>)).

Advice on exercise and physical activity

- **Advise adults with type 1 diabetes that regular exercise:**
 - May lower blood glucose levels, as exercise increases the amount of glucose used by the muscles for energy.
 - Can reduce the increased cardiovascular risk in the medium and longer term.
 - Can help with weight management (when combined with a healthy diet). See the CKS topic on [Obesity](#) ([/topics/obesity/](#)) for more information.
- **Ensure the person has information on:**
 - The role of [self-monitoring](#) ([/topics/diabetes-type-1/management/management-adults/#self-monitoring-of-blood-glucose](#)) on changed insulin and/or nutritional needs.
 - The effect of activity on blood glucose levels when insulin levels are adequate (risk of hypoglycaemia).
 - The effect of exercise on blood glucose levels when the person is hyperglycaemic or hypoinsulinaemic (risk of worsening of hyperglycaemia and ketonaemia).
 - The appropriate adjustments of insulin dosage and/or nutritional intake for exercise and post-exercise periods, and the next 24 hours.
 - Interactions of exercise and alcohol.
 - Further contacts and sources of information. Information on [Diabetes and exercise](#) (<https://www.diabetes.org.uk/Guide-to-diabetes/Managing-your-diabetes/Exercise>) is available on the Diabetes UK website (www.diabetes.org.uk) (<https://www.diabetes.org.uk>)).
- **For people who choose to integrate increased physical activity into a more healthy lifestyle, provide information on the appropriate intensity and frequency.** The current [UK Chief Medical Officers' Physical Activity Guidelines](#) (https://assets.publishing.service.gov.uk/government/uploads/system/uploads/attachment_data/file/832868/uk-chief-medical-officers-physical-activity-guidelines.pdf) state that:
 - **Adults aged 19–64 years should:**
 - Aim to be physically active every day for good physical and mental health.
 - Do activities to develop or maintain strength in the major muscle groups. These could include heavy gardening, carrying heavy shopping, or resistance exercise. Muscle strengthening activities should be done on at least two days a week, but any strengthening activity is better than none.
 - Each week, accumulate at least 150 minutes of moderate intensity activity (such as brisk walking or cycling), or 75 minutes of vigorous intensity activity (such as running), or shorter durations of very vigorous intensity activity (such as sprinting or stair climbing), or a combination of moderate, vigorous, and very vigorous intensity activity.
 - Aim to minimize the amount of time spent being sedentary and when physically possible should break up long periods of inactivity with at least light physical activity.
 - **Older adults aged 65 years and over should:**

- Participate in daily physical activity to gain health benefits, including maintenance of good physical and mental health, wellbeing, and social functioning. Some physical activity is better than none; even light activity brings some health benefits compared with being sedentary, while more daily physical activity provides greater health and social benefits.
- Maintain or improve their physical function by undertaking activities aimed at improving or maintaining muscle strength, balance, and flexibility on at least two days a week. These could be combined with sessions involving moderate aerobic activity or could be additional sessions aimed specifically at these components of fitness.
- Each week, aim to accumulate 150 minutes of moderate intensity aerobic activity, building up gradually from current levels. Those who are already regularly active can achieve these benefits through 75 minutes of vigorous intensity activity or a combination of moderate and vigorous activity to achieve greater benefits. Weight-bearing activities that create an impact through the body help to maintain bone health.
- Break up prolonged periods of being sedentary with light activity when physically possible, or at least with standing, as this has distinct health benefits for older people.

Advice on smoking and substance misuse

- **Explain that smoking is a risk factor for cardiovascular disease.**
 - Give adults with type 1 diabetes who smoke advice on smoking cessation and the use of smoking cessation services.
 - Reinforce this advice annually for people who currently do not plan to stop smoking, and at all clinical contacts if there is a prospect of the person stopping.
 - Advise young adult non-smokers never to start smoking.
 - See the CKS topic on [Smoking cessation \(/topics/smoking-cessation/\)](/topics/smoking-cessation/) for information on smoking cessation and smoking cessation services.
- **Explain about the general dangers of substance misuse and the possible effects on blood glucose control.**
 - Information, advice, and support on substance misuse are available on the Talk to Frank website ([www.talktofrank.com \(http://www.talktofrank.com/\)](http://www.talktofrank.com/)).

Advice on oral health

- Advise adults with type 1 diabetes at their annual review of self-care and needs:
 - That they are at higher risk of periodontitis.
 - That if they get periodontitis, managing it can improve their blood glucose control and can reduce their risk of hyperglycaemia.
 - To have regular oral health reviews (their oral healthcare or dental team will tell them how often this is required).
- For adults with type 1 diabetes who have been diagnosed with periodontitis by an oral healthcare or dental team, they should be advised to attend appointments to manage and treat their periodontitis (at a frequency based on their oral health needs).

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How should I manage an adult with hypoglycaemia?

- See the section on [Hypoglycaemia \(/topics/insulin-therapy-in-type-1-diabetes/management/insulin-therapy-type-1-diabetes/#managing-hypoglycaemia\)](/topics/insulin-therapy-in-type-1-diabetes/management/insulin-therapy-type-1-diabetes/#managing-hypoglycaemia) in the CKS topic on [Insulin therapy in type 1 diabetes \(/topics/insulin-therapy-in-type-1-diabetes/\)](/topics/insulin-therapy-in-type-1-diabetes/) for management information.

How should I manage an adult with diabetic ketoacidosis

- If diabetic ketoacidosis (DKA) is **suspected** (</topics/diabetes-type-1/diagnosis/when-to-suspect-diabetic-ketoacidosis/>), admit the person immediately for confirmation of the diagnosis and emergency treatment with intravenous insulin and fluids.
- After the person has recovered from an episode of DKA:
 - Discuss the [factors \(/topics/diabetes-type-1/diagnosis/when-to-suspect-diabetic-ketoacidosis/\)](/topics/diabetes-type-1/diagnosis/when-to-suspect-diabetic-ketoacidosis/) that may have led to the episode.
 - If DKA is recurrent, consider the possibility of non-adherence to treatment or intentional insulin omission in order to lose weight ([diabulimia \(/topics/diabetes-type-1/background-information/complications/\)](/topics/diabetes-type-1/background-information/complications/)).
 - Give general information on DKA, including information on how to reduce the risk of future episodes.
 - Information on DKA for people with type 1 diabetes and their family/carer is available on the Diabetes UK website (www.diabetes.org.uk (https://www.diabetes.org.uk/guide-to-diabetes/complications/diabetic_ketoacidosis/)).
 - Ensure that the person is given appropriate information on how to manage diabetes during a period of illness ('sick-day rules' (</topics/diabetes-type-1/management/management-adults/#sick-day-rules/>)).

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How should I manage an adult with type 1 diabetes during illness?

- **Consider the need for admission or specialist advice.** The decision will depend on clinical judgement, taking into account the person's age, any underlying illness or complications, and the presence of hyperglycaemia (random plasma glucose more than 11 mmol/L) and ketosis.
 - **Arrange hospital admission if:**
 - There is an immediate risk of [diabetic ketoacidosis \(/topics/diabetes-type-1/diagnosis/when-to-suspect-diabetic-ketoacidosis/\)](/topics/diabetes-type-1/diagnosis/when-to-suspect-diabetic-ketoacidosis/) (DKA).
 - There is moderate ketonuria (2+ on urine dipstick) or ketonaemia (1.5–2.9 mmol/L) with or without hyperglycaemia and the person cannot eat or drink, as there is a risk of DKA.
 - The person does not improve rapidly with insulin treatment.
 - **Consider arranging hospital admission or seeking urgent specialist advice if:**
 - The underlying condition is unclear.
 - The person is dehydrated or at risk of dehydration.
 - Vomiting persists beyond 2 hours.

- The person and their family/carers are unable to keep the blood glucose level above 3.5 mmol/L.
- The person is on continuous subcutaneous insulin pump therapy. See the section on [Insulin pump therapy \(/topics/insulin-therapy-in-type-1-diabetes/management/insulin-therapy-type-1-diabetes/#recommended-insulin-regimen\)](/topics/insulin-therapy-in-type-1-diabetes/management/insulin-therapy-type-1-diabetes/#recommended-insulin-regimen) in the CKS topic on [Insulin therapy in type 1 diabetes \(/topics/insulin-therapy-in-type-1-diabetes/\)](/topics/insulin-therapy-in-type-1-diabetes/).
- The person and their family/carers are exhausted, for example due to repeated night-time waking.
- Language problems make it difficult to communicate with the person and/or their family/carers.
- **If the person can be managed in primary care:**
 - Assess and manage the intercurrent illness.
 - Assess how well blood glucose has been monitored and managed with diet and insulin.
 - Warn the person and their family/carers that illness may affect blood glucose control:
 - Some illnesses (especially if associated with fever) can raise blood glucose levels.
 - In contrast, illnesses associated with vomiting and diarrhoea (such as gastroenteritis) can lower blood glucose levels, possibly leading to hypoglycaemia (blood glucose levels less than 3.5 mmol/L).
 - Ensure that the person has:
 - Written 'sick-day rules' (</topics/diabetes-type-1/management/management-adults/#sick-day-rules>) and contact details of the healthcare team.
 - Easily digestible food and sugary drinks (to provide energy and to prevent further ketosis).
 - Oral rehydration salt sachets (to prevent dehydration).
 - Additional supplies of insulin.
 - Glucose tablets or oral gel (to prevent hypoglycaemia).
 - A glucagon kit (if appropriate).

What are the 'sick-day rules' for adults with type 1 diabetes?

- **During a period of illness that does not require admission, remind the person to adhere to the 'sick-day rules' that should have been provided by their diabetes team.**
They should:
 - **Never stop or omit insulin.**
 - The dose of insulin may need to be altered during periods of illness; they should seek advice from their diabetes team if they are unsure of how to adjust insulin doses.
 - **Check blood glucose more frequently, for example every 1–2 hours including through the night.**
 - The insulin dose should be titrated according to the blood glucose results and the written 'sick-day rules'.
 - **Consider checking blood or urine ketone levels regularly,** for example every 3–4 hours including through the night, and sometimes every 1–2 hours depending on results.
 - If the urine ketone level is greater than 2+ or blood ketone levels are greater than 3 mmol/L, the person should contact the GP or diabetes care team immediately.
 - **Maintain their normal meal pattern (where possible) if appetite is reduced.**
 - Their normal meals could be replaced with carbohydrate-containing drinks (such as milk, milk shakes, fruit juices, and sugary drinks).

- **Aim to drink at least 3 L of fluid (5 pints) a day to prevent dehydration.**
 - If blood glucose levels are normal or high, water or sugar-free fluids are probably most appropriate in the majority of cases.
 - If blood glucose levels are low, drinks containing glucose are required (or the person should take carbohydrates if possible).
 - Carbonated drinks should be avoided if possible.
- **Seek urgent medical advice if they are violently sick, drowsy, or unable to keep fluids down.**
 - Intravenous fluids may be required.
- **When feeling better, continue to monitor their blood glucose carefully until it returns to normal.**
 - This may take some time.
 - The person should seek medical advice if their blood glucose remains uncontrolled.
- **Further patient information on managing diabetes during a period of illness is available on the Diabetes UK website (www.diabetes.org.uk (https://www.diabetes.org.uk/Guide-to-diabetes/Living_with_diabetes/Illness/)).**

Basis for recommendation

▼ [Show](#)

How should I manage cardiovascular risk factors in adult diabetes?

- **Assess the following cardiovascular risk factors annually:**
 - **Lifestyle, including smoking status.**
 - See the sections on [Diet \(/topics/diabetes-type-1/management/management-adults/#diet\)](/topics/diabetes-type-1/management/management-adults/#diet), [Exercise \(/topics/diabetes-type-1/management/management-adults/#exercise-physical-activity\)](/topics/diabetes-type-1/management/management-adults/#exercise-physical-activity), [Alcohol intake \(/topics/diabetes-type-1/management/management-adults/#alcohol-intake\)](/topics/diabetes-type-1/management/management-adults/#alcohol-intake), and [Substance misuse \(/topics/diabetes-type-1/management/management-adults/#smoking-substance-misuse\)](/topics/diabetes-type-1/management/management-adults/#smoking-substance-misuse) for information on managing these lifestyle issues.
 - **Waist circumference.**
 - The waist circumference provides an indication of total body fat and a risk of obesity-related health problems.
 - See the CKS topic on [Obesity \(/topics/obesity/\)](/topics/obesity/) for more information.
 - **Blood glucose control.**
 - Advise on the importance of adequate blood glucose control.
 - Reiterate [HbA1c \(/topics/diabetes-type-1/management/management-adults/#target-hba1c\)](/topics/diabetes-type-1/management/management-adults/#target-hba1c) and [glucose self-monitoring \(/topics/diabetes-type-1/management/management-adults/#self-monitoring-of-blood-glucose\)](/topics/diabetes-type-1/management/management-adults/#self-monitoring-of-blood-glucose) targets.
 - **Blood pressure.**
 - See the section on [Antihypertensive treatment \(/topics/diabetes-type-1/management/management-adults/#antihypertensive-treatment\)](/topics/diabetes-type-1/management/management-adults/#antihypertensive-treatment) for more information.
 - **Albuminuria.**
 - See the section on [Diabetic kidney disease \(/topics/diabetes-type-1/management/management-adults/#diabetic-kidney-disease\)](/topics/diabetes-type-1/management/management-adults/#diabetic-kidney-disease) for more information.

- **Full lipid profile** (high-density lipoprotein [HDL] cholesterol, low-density lipoprotein [LDL] cholesterol, and triglycerides).
 - Offer a statin for the prevention of cardiovascular disease (CVD).
 - See the section on [Lipid modification therapy \(/topics/diabetes-type-1/management/management-adults/#lipid-modification\)](/topics/diabetes-type-1/management/management-adults/#lipid-modification) for more information.
- **Family history of CVD.**
- **Provide intensive management for adults who have had myocardial infarction or stroke.**
 - Consider a beta blocker in people with angina or other ischaemic heart disease. See the CKS topics on [MI - secondary prevention \(/topics/mi-secondary-prevention/\)](/topics/mi-secondary-prevention/) and [Stroke and TIA \(/topics/stroke-tia/\)](/topics/stroke-tia/) for more information.
 - **Do not offer aspirin for the primary prevention of CVD.**
 - Decisions about antiplatelet treatment will usually be made by, or in consultation with, secondary care.
 - See the CKS topic on [Antiplatelet treatment \(/topics/antiplatelet-treatment/\)](/topics/antiplatelet-treatment/) for information on the use of aspirin for the secondary prevention of CVD.

Antihypertensive treatment

- **Aim for blood pressure targets as follows:**
 - For adults with a urine albumin:creatinine ratio (ACR) less than 70 mg/mmol, aim for a clinic systolic blood pressure less than 140 mmHg (target range 120 to 139 mmHg) and a clinic diastolic blood pressure less than 90 mmHg.
 - For adults with an ACR of 70 mg/mmol or more, aim for a clinic systolic blood pressure less than 130 mmHg (target range 120 to 129 mmHg) and a clinic diastolic blood pressure less than 80 mmHg.
 - In adults aged 80 or more, whatever the ACR, aim for a clinic systolic blood pressure less than 150 mmHg (target range 140 to 149 mmHg) and a clinic diastolic blood pressure less than 90 mmHg.
- Use clinical judgement for adults with frailty, target organ damage (damage to organs because of diabetes, for example, to nerves or eyes) or multimorbidity.
- **Start a trial of a renin-angiotensin system blocking drug as first-line treatment for hypertension in adults with type 1 diabetes.**
 - Offer an angiotensin-converting enzyme (ACE) inhibitor, provided there are no contraindications. If an ACE inhibitor is not tolerated, offer an angiotensin-II receptor antagonist (AIIA) if appropriate.
 - Start with a low dose and titrate up to the maximum tolerated therapeutic dose (within the maximum licensed dose) by doubling the dose every 1–2 weeks.
 - After each upward titration, monitor the person's renal function, serum potassium level, and blood pressure.
 - See the section on [Prescribing information \(/topics/hypertension/prescribing-information/\)](/topics/hypertension/prescribing-information/) in the CKS topic on [Hypertension - not diabetic \(/topics/hypertension/\)](/topics/hypertension/) for detailed information on prescribing ACE inhibitors and AIIAs.
 - Provide information on the potential for lifestyle changes to improve blood pressure control and associated outcomes. See the sections on [Smoking \(/topics/diabetes-type-1/management/management-adults/#smoking-substance-misuse/\)](/topics/diabetes-type-1/management/management-adults/#smoking-substance-misuse/), [Diet \(/topics/diabetes-type-1/management/management-adults/#diet/\)](/topics/diabetes-type-1/management/management-adults/#diet/), and [Exercise \(/topics/diabetes-type-1/management/management-adults/#exercise-physical-activity/\)](/topics/diabetes-type-1/management/management-adults/#exercise-physical-activity/) for more information.
 - Be aware that it may be necessary to prescribe other antihypertensive drugs to improve blood pressure control.

- Do not avoid cardioselective beta-blockers (such as atenolol and metoprolol) where indicated in adults on insulin.
- Low-dose thiazides may be combined with beta-blockers.
- When calcium-channel blockers are prescribed, use only long-acting preparations.
- For information on managing hypertension in adults with type 1 diabetes and nephropathy, see the section on [Diabetic kidney disease \(/topics/diabetes-type-1/management/management-adults/#diabetic-kidney-disease\)](/topics/diabetes-type-1/management/management-adults/#diabetic-kidney-disease).

Lipid modification therapy

Do not use a risk assessment tool to assess cardiovascular disease (CVD) risk in people with type 1 diabetes.

- **For people with type 1 diabetes who do not have established CVD** (for example a history of myocardial infarction, angina, stroke, transient ischaemic attack, or peripheral arterial disease):
 - Offer statin treatment with atorvastatin 20 mg for the primary prevention of CVD if the person:
 - Aged over 40 years.
 - Has had diabetes for more than 10 years.
 - Has established nephropathy.
 - Has other CVD risk factors (such as obesity and hypertension).
 - For all other adults with type 1 diabetes, consider statin treatment with atorvastatin 20 mg for the primary prevention of CVD.
- **For people with type 1 diabetes who have established CVD:**
 - Advise statin treatment with atorvastatin 80 mg for the secondary prevention of CVD.
- **Before initiating treatment with a statin:**
 - Implement measures to reduce the risk of CVD. These are discussed in detail in the CKS topic on [CVD risk assessment and management \(/topics/cvd-risk-assessment-management/\)](/topics/cvd-risk-assessment-management/), and include:
 - Addressing other modifiable CVD risk factors, such as smoking.
 - Identifying and managing secondary causes of hyperlipidaemia, such as excess alcohol consumption, hypothyroidism, liver disease, and nephrotic syndrome.
 - Have an informed discussion with the person and/or their family/carers about the risks and benefits of statin treatment, taking into account additional factors, such as potential benefits from lifestyle modifications, the person's preference, comorbidities, polypharmacy, general frailty, and life expectancy. Advise that:
 - Statin treatment reduces CVD risk compared with no treatment.
 - The adverse effects of statins are generally mild, reversible, and not medically serious, although some people may find them distressing.
 - Myopathy and rhabdomyolysis are the most serious adverse effects of statins. However, they are rare, with estimated incidences of 5 cases per 100,000 person-years and 1.6 cases per 100,000 person-years, respectively.
 - Perform baseline blood tests to ensure that statin treatment is suitable. This should include:
 - A non-fasting lipid profile, liver function tests, and renal function.
 - Thyroid stimulating hormone — if hyperlipidaemia is present.
 - Creatine kinase — if the person has persistent generalized unexplained muscle pain.

- See the CKS topic on [Lipid modification - CVD prevention \(/topics/lipid-modification-cvd-prevention/\)](/topics/lipid-modification-cvd-prevention/) for detailed information on the baseline tests, how to initiate statin treatment, and how to monitor a person taking a statin.

Basis for recommendation

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How should I monitor for complications in adults with

People with type 1 diabetes should have regular surveillance for [complications \(/topics/diabetes-type-1/background-information/complications/\)](/topics/diabetes-type-1/background-information/complications/) of the disease. The reason for the surveillance as well as the significance of any abnormal findings should be discussed with the person and their family/carers.

- **At every review appointment (if clinically indicated):**
 - Measure HbA1c to ensure adequate blood glucose control. Reiterate [HbA1c \(/topics/diabetes-type-1/management/management-adults/#target-hba1c\)](/topics/diabetes-type-1/management/management-adults/#target-hba1c) and [glucose self-monitoring \(/topics/diabetes-type-1/management/management-adults/#target-glucose-self-monitoring-levels\)](/topics/diabetes-type-1/management/management-adults/#target-glucose-self-monitoring-levels) targets.
 - Measure the person's height, weight, and waist circumference, and calculate their body mass index (BMI). See the CKS topic on [Obesity \(/topics/obesity/\)](/topics/obesity/) for information on managing obesity.
 - Assess for depression, anxiety, and eating disorders (including intentional insulin omission in order to lose weight [[diabulimia \(/topics/diabetes-type-1/background-information/complications/\)](/topics/diabetes-type-1/background-information/complications/)]).
 - Alerting signs of an eating disorder include over-concern with body shape and weight, low BMI, hypoglycaemia, and suboptimal overall blood glucose control.
 - See the CKS topics on [Depression \(/topics/depression/\)](/topics/depression/), [Generalized anxiety disorder \(/topics/generalized-anxiety-disorder/\)](/topics/generalized-anxiety-disorder/), and [Eating disorders \(/topics/eating-disorders/\)](/topics/eating-disorders/) for information on managing these conditions.
 - Check smoking status. If applicable, offer advice on smoking cessation. See the CKS topic on [Smoking cessation \(/topics/smoking-cessation/\)](/topics/smoking-cessation/) for more information.
 - Monitor for neuropathy and associated complications, including erectile dysfunction, neuropathic pain, autonomic neuropathy, and gastroparesis (delayed emptying of the stomach). See the section on [Managing neuropathic complications \(/topics/diabetes-type-1/management/management-adults/#neuropathy\)](/topics/diabetes-type-1/management/management-adults/#neuropathy) if any neuropathic complications are identified.
- **Once a year:**
 - Check injection sites, and address any injection site problems. See the section on [Managing injection site problems \(/topics/insulin-therapy-in-type-1-diabetes/management/insulin-therapy-type-1-diabetes/#managing-injection-site-problems\)](/topics/insulin-therapy-in-type-1-diabetes/management/insulin-therapy-type-1-diabetes/#managing-injection-site-problems) in the CKS topic on [Insulin therapy in type 1 diabetes \(/topics/insulin-therapy-in-type-1-diabetes/\)](/topics/insulin-therapy-in-type-1-diabetes/) for more information.
 - Assess for cardiovascular risk factors, including smoking status, waist circumference, blood glucose control, blood pressure, full lipid profile, and family history of cardiovascular disease. See the section on [Managing cardiovascular risk factors \(/topics/diabetes-type-1/management/management-adults/#managing-cardiovascular-risk\)](/topics/diabetes-type-1/management/management-adults/#managing-cardiovascular-risk) for more information.
 - Ensure the person is screened for eye disease, kidney disease, and foot problems. See the sections on [Eye disease \(/topics/diabetes-type-1/management/management-adults/#eye-disease\)](/topics/diabetes-type-1/management/management-adults/#eye-disease), [Diabetic kidney disease \(/topics/diabetes-type-1/management/management-adults/#diabetic-kidney-](/topics/diabetes-type-1/management/management-adults/#diabetic-kidney-disease)

- disease), and [Diabetic foot problems \(/topics/diabetes-type-1/management/management-adults/#foot-problems\)](#) for more information.
 - Ensure the person is screened for thyroid disease. See the CKS topics on [Hypothyroidism \(/topics/hypothyroidism/\)](#) and [Hyperthyroidism \(/topics/hyperthyroidism/\)](#) for detailed information on screening for (and managing) these conditions.
- **Be alert for the possibility of the development of other autoimmune diseases, including:**
 - Coeliac disease — this should have been screened for at diagnosis.
 - Assess for coeliac disease if the person has a low BMI or unexplained weight loss. See the CKS topic on [Coeliac disease \(/topics/coeliac-disease/\)](#) for more information.
 - Addison's disease. See the section on [Diagnosis \(/topics/addisons-disease/diagnosis/\)](#) in the CKS topic on [Addison's disease \(/topics/addisons-disease/\)](#) for information on the clinical features of Addison's disease.
 - Pernicious anaemia. See the section on [Diagnosis \(/topics/anaemia-b12-folate-deficiency/diagnosis/\)](#) in the CKS topic on [Anaemia - B12 and folate deficiency \(/topics/anaemia-b12-folate-deficiency/\)](#) for information on the clinical features of pernicious anaemia.

Screening for eye disease

- **On diagnosis, immediately refer adults with type 1 diabetes to the local eye screening service.**
 - Perform screening as soon as possible and no later than 3 months from referral.
 - Arrange repeat structured eye screening annually.
- **Depending on the findings, follow structured eye screening by one of the following:**
 - Referral to an ophthalmologist.
 - Earlier review.
 - Routine annual review.
 - The [NHS diabetic eye screening \(https://www.gov.uk/guidance/diabetic-eye-screening-programme-overview\)](https://www.gov.uk/guidance/diabetic-eye-screening-programme-overview) (DES) is offered once a year to anyone with diabetes who is aged 12 years or over.
 - The programme aims to reduce the risk of sight loss among people with diabetes by the early detection and treatment, if needed, of sight-threatening retinopathy.
- **Explain the reasons and success of eye screening systems to the person and their family/carers** so that attendance is not reduced by lack of knowledge or fear of outcome.
- **Arrange:**
 - **Emergency ophthalmology review if there is:**
 - Sudden loss of vision.
 - Rubeosis iridis (formation of abnormal blood vessels on the anterior iris).
 - Pre-retinal or vitreous haemorrhage.
 - Retinal detachment.
 - **Urgent ophthalmology review if there is:**
 - Formation of new abnormal vessels on the retina.

Diabetic kidney disease

- **All adults with type 1 diabetes (with or without detected nephropathy) should receive annual screening for diabetic nephropathy.**

- **Ask the person to bring in the first urine sample of the day ('early morning urine') once a year.**
 - Send the urine specimen for estimation of the albumin:creatinine ratio (ACR).
 - Check serum creatinine at the same time to calculate estimated glomerular filtration rate (eGFR).
- **Diagnose chronic kidney disease (CKD)** if the person's eGFR is persistently less than 60 mL per minute per 1.73 m² and/or ACR is persistently greater than 3 mg/mmol.
 - See the section on [Diagnosis \(/topics/chronic-kidney-disease/diagnosis/\)](/topics/chronic-kidney-disease/diagnosis/) in the CKS topic on [Chronic kidney disease \(/topics/chronic-kidney-disease/\)](/topics/chronic-kidney-disease/) for more information on diagnosing CKD.
- **Suspect other renal disease:**
 - In the absence of progressive retinopathy.
 - If blood pressure is particularly high.
 - If proteinuria develops suddenly.
 - If the person is systemically unwell.
 - If significant persistent non-visible haematuria is present (that is, two out of three dipstick tests show 1+ or more of blood after exclusion of a urinary tract infection).
- **For adults with type 1 diabetes who have confirmed nephropathy, including those with moderately increased albuminuria (microalbuminuria) alone:**
 - Consider whether referral to a nephrology specialist is indicated, for example if the person has an eGFR of less than 30 mL per minute per 1.73 m². See the section on [Referral \(/topics/chronic-kidney-disease/management/management-of-chronic-kidney-disease/#referral\)](/topics/chronic-kidney-disease/management/management-of-chronic-kidney-disease/#referral) in the CKS topic on [Chronic kidney disease \(/topics/chronic-kidney-disease/\)](/topics/chronic-kidney-disease/) for information on when to arrange referral to a nephrology specialist.
 - Offer an angiotensin-converting enzyme (ACE) inhibitor provided there are no contraindications. If ACE inhibitors are contraindicated or not tolerated, offer an angiotensin-II receptor antagonist (AIIRA) if appropriate.
 - Start with a low dose and titrate up to the maximum tolerated therapeutic dose (within the maximum licensed dose), by doubling the dose every 1–2 weeks. After each upward titration, monitor the person's renal function, serum potassium level, and blood pressure.
 - Do not offer a combination of an ACE inhibitor and an AIIRA.
 - See the section on [Prescribing information \(/topics/chronic-kidney-disease/prescribing-information/\)](/topics/chronic-kidney-disease/prescribing-information/) in the CKS topic on [Chronic kidney disease \(/topics/chronic-kidney-disease/\)](/topics/chronic-kidney-disease/) for detailed information on prescribing an ACE inhibitor or AIIRA.
 - See the section on [Managing Cardiovascular risk \(/topics/diabetes-type-1/management/management-adults/#managing-cardiovascular-risk\)](/topics/diabetes-type-1/management/management-adults/#managing-cardiovascular-risk) for target blood pressures.
 - Advise the person to avoid a high protein diet. Consider referring the person to a dietitian or the specialist diabetes team for dietary advice.
 - Offer lipid modification therapy (without the need for a formal risk assessment). See the CKS topic on [Lipid modification - CVD prevention \(/topics/lipid-modification-cvd-prevention/\)](/topics/lipid-modification-cvd-prevention/) for more information.
 - See the section on [Primary care management \(/topics/chronic-kidney-disease/management/management-of-chronic-kidney-disease/#management-in-primary-care\)](/topics/chronic-kidney-disease/management/management-of-chronic-kidney-disease/#management-in-primary-care) in the CKS topic on [Chronic kidney disease \(/topics/chronic-kidney-disease/\)](/topics/chronic-kidney-disease/) for further management information.

Neuropathy

Chronic painful neuropathy

- See the CKS topic on [Neuropathic pain - drug treatment \(/topics/neuropathic-pain-drug-treatment/\)](/topics/neuropathic-pain-drug-treatment/) for information on drug treatments for neuropathic pain.

Erectile dysfunction

- Offer men the opportunity to discuss erectile dysfunction as part of their regular review.
- Offer a phosphodiesterase-5 (PDE-5) inhibitor (sildenafil, vardenafil, or tadalafil) to men with type 1 diabetes with isolated erectile dysfunction, unless contraindicated. Choose the PDE-5 inhibitor with the lowest acquisition cost. See the section on [Prescribing information \(/topics/erectile-dysfunction/prescribing-information/\)](/topics/erectile-dysfunction/prescribing-information/) in the CKS topic on [Erectile dysfunction \(/topics/erectile-dysfunction/\)](/topics/erectile-dysfunction/) for detailed information on prescribing PDE-5 inhibitors, including contraindications and cautions.
- If PDE-5 inhibitor treatment is contraindicated or unsuccessful, consider referring the man to a service offering further assessment and other medical, surgical, or psychological management of erectile dysfunction.

Autonomic neuropathy

- For adults with unexplained diarrhoea, particularly at night, consider (and investigate) the possibility of autonomic neuropathy affecting the gut.
- For adults with bladder emptying problems, consider (and investigate) the possibility of autonomic neuropathy affecting the bladder, provided alternative causes (for example lower urinary tract problems) have been excluded. See the CKS topics on [LUTS in men \(/topics/luts-in-men/\)](/topics/luts-in-men/), [Urinary tract infection \(lower\) - men \(/topics/urinary-tract-infection-lower-men/\)](/topics/urinary-tract-infection-lower-men/), and [Urinary tract infection \(lower\) - women \(/topics/urinary-tract-infection-lower-women/\)](/topics/urinary-tract-infection-lower-women/) for more information on managing these conditions.
- For adults with vomiting caused by gastroparesis:
 - Advise a small-particle-size diet (mashed or pureed food) for symptomatic relief.
 - Consider referral to a gastroenterologist or the diabetes specialist team for consideration of treatment with a prokinetic drug, such as erythromycin (off-label use), metoclopramide, or domperidone.
 - Continuous subcutaneous insulin infusion (CSII or insulin pump) therapy may be considered in secondary care if there are ongoing symptoms.
 - If symptoms persist despite insulin pump therapy, further specialist referral is likely to be needed.
 - See the section on [Recommended insulin regimen \(/topics/insulin-therapy-in-type-1-diabetes/management/insulin-therapy-type-1-diabetes/#recommended-insulin-regimen\)](/topics/insulin-therapy-in-type-1-diabetes/management/insulin-therapy-type-1-diabetes/#recommended-insulin-regimen) in the CKS topic on [Insulin therapy in type 1 diabetes \(/topics/insulin-therapy-in-type-1-diabetes/\)](/topics/insulin-therapy-in-type-1-diabetes/) for information on insulin pump therapy.
- For adults with excessive sweating, see the CKS topic on [Hyperhidrosis \(/topics/hyperhidrosis/\)](/topics/hyperhidrosis/).
- For adults taking antihypertensive drugs, be aware of the increased risks of postural hypotension as a result of the combined effects of sympathetic autonomic neuropathy and antihypertensive drug treatment.

Acute painful neuropathy of rapid improvement of blood glucose control

- Reassure the person that acute painful neuropathy resulting from rapid improvement of blood glucose control is a self-limiting condition that improves symptomatically over time.

- Explain that the specific treatments for this condition:
 - Aim to make the symptoms tolerable until the condition resolves.
 - May not relieve pain immediately and may need to be taken regularly for several weeks to be effective.
- Advise the use of simple analgesics (paracetamol or a nonsteroidal anti-inflammatory drug [NSAID]) and local measures (such as a bed cradle — a frame that attaches to the bed to keep sheets and blankets from touching and rubbing the body) to relieve symptoms.
 - If simple analgesics are ineffective, offer treatment as described in the CKS topic on [Neuropathic pain - drug treatment \(/topics/neuropathic-pain-drug-treatment/\)](/topics/neuropathic-pain-drug-treatment/).
 - Simple analgesia may be continued until the effects of additional treatments have been established.
 - The use of weak opioids is not recommended owing to the risk of dependency associated with their use.
- Do not relax blood glucose diabetes control if this condition is identified.

Screening for diabetic foot problems

People with type 1 diabetes should have their feet checked by a primary healthcare professional at diagnosis and at least once a year thereafter, or sooner if any foot problems arise. They should also be advised to check their own feet on a daily basis.

- **To examine the feet of a person with diabetes, remove their shoes, socks, bandages, and dressings (as appropriate) and examine both feet for evidence of the following risk factors:**
 - Neuropathy — use a 10 g monofilament as part of a foot sensory examination.
 - Limb ischaemia — use ankle brachial pressure index (ABPI) and interpret results carefully because calcified arteries may falsely elevate results. See the CKS topic on [Peripheral arterial disease \(/topics/peripheral-arterial-disease/\)](/topics/peripheral-arterial-disease/) for more information.
 - Ulceration.
 - Callus.
 - Infection and/or inflammation.
 - Deformity.
 - Gangrene.
 - Charcot arthropathy (an acute, localized inflammatory condition that may lead to varying degrees and patterns of bone destruction, subluxation, dislocation, and deformity).
- **Assess the person's current risk of developing a diabetic foot problem or needing an amputation using the following risk stratification:**
 - Low risk:
 - No risk factors present except callus alone.
 - Moderate risk:
 - Deformity, *or*
 - Neuropathy, *or*
 - Peripheral arterial disease.
 - High risk:
 - Previous ulceration, *or*
 - Previous amputation, *or*
 - On renal replacement therapy, *or*
 - Neuropathy and peripheral arterial disease together, *or*
 - Neuropathy in combination with callus and/or deformity, *or*

- Peripheral arterial disease in combination with callus and/or deformity.
- Active diabetic foot problem:
 - Ulceration, *or*
 - Infection, *or*
 - Critical limb-threatening ischaemia, *or*
 - Gangrene, *or*
 - Suspicion of an acute Charcot arthropathy, or an unexplained hot, swollen foot with a change in colour, with or without pain.
- **Manage the person's risk of developing a diabetic foot problem.**
 - For people who are at low risk:
 - Continue to carry out annual foot assessments as part of their annual diabetes review.
 - Emphasize the importance of foot care.
 - Advise that they could progress to moderate or high risk.
 - For people at moderate risk — refer to the foot protection service to be seen within 6–8 weeks.
 - For people at high risk — refer to the foot protection service to be seen within 2–4 weeks.
 - For people with a limb-threatening or life-threatening diabetic foot problem, refer immediately to acute services, and inform the multidisciplinary foot care service (according to local protocols and pathways) so that they can be assessed and an individualized treatment plan put in place. Examples of limb-threatening and life-threatening diabetic foot problems include:
 - Ulceration with fever or any signs of sepsis.
 - Ulceration with limb ischaemia.
 - Clinical concern that there is a deep-seated soft tissue or bone infection (with or without ulceration).
 - Gangrene (with or without ulceration).
 - For all other active diabetic foot problems, refer the person within 1 working day to the multidisciplinary foot care service or foot protection service (according to local protocols and pathways) for triage within 1 further working day.
- **Depending on the person's risk of developing a diabetic foot problem, carry out reassessments at the following intervals:**
 - Annually — for people who are at low risk, as part of their annual diabetes review.
 - Frequently (for example every 3–6 months) — for people who are at moderate risk.
 - More frequently (for example every 1–2 months) — for people who are at high risk, if there is no immediate concern.
 - Very frequently (for example every 1–2 weeks) — for people who are at high risk, if there is immediate concern.
- Consider more frequent reassessments for people who are at moderate or high risk, and for people who are unable to check their own feet.
- **Provide appropriate information on the risk of developing a diabetic foot problem to all people with type 1 diabetes.** This information should be provided at diagnosis, during assessments, and if foot problems arise, and should include the following:
 - Basic foot care advice and the importance of foot care.
 - Foot emergencies and who to contact.
 - Footwear advice.
 - The person's current individual risk of developing a foot problem.
 - Information on diabetes and the importance of blood glucose control.

- If a person presents with a diabetic foot problem, take into account that they may have an undiagnosed, increased risk of cardiovascular disease (CVD) that may need further investigation and treatment. See the CKS topics on [CVD risk assessment and management \(/topics/cvd-risk-assessment-management/\)](/topics/cvd-risk-assessment-management/) and [Lipid modification - CVD prevention \(/topics/lipid-modification-cvd-prevention/\)](/topics/lipid-modification-cvd-prevention/) for more information on the investigation and treatment of CVD.

Basis for recommendation

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